

158.888 Information Technology Professional Project

**Reflection and Peer Review Report on
Real-Time Parking Occupancy Monitoring and Area-
Level Prediction Using Spatio-Temporal Analytics**

Presented by
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1. Reflection on the Project

This project enhanced my understanding of predictive analytics, machine learning workflows, and intelligent transportation systems. Through participation in the project, I gained practical experience in data pre-processing, feature engineering, machine learning model implementation, and analytical visualisation techniques. The project further improved my technical understanding of how predictive technologies can support smart city and transportation management applications.

One of the most interesting aspects of the project was the integration of GIS-based hotspot visualization with operational parking analytics. The project demonstrated how geographic intelligence and occupancy forecasting can be combined to support urban transportation analysis and parking management decision-making. Working with hotspot mapping and occupancy categorization significantly improved my understanding of spatial analytics and smart city visualization techniques.

The project also provided practical experience in integrating machine learning outputs with GIS-based visualization frameworks and operational dashboards. Observing how predictive analytics can be transformed into meaningful spatial intelligence visualizations helped improve my understanding of operational transportation analytics systems.

Several practical challenges were encountered during implementation including inconsistent location naming, varying occupancy structures, real-time parking feed integration, and dashboard testing. Participating in these implementations and testing activities significantly improved my understanding of transportation analytics workflows and operational reporting systems. Debugging occupancy categorization issues, refining hotspot visualizations, and validating dashboard behaviour provided highly valuable technical experience.

Another important learning outcome was understanding the importance of data quality and preprocessing in predictive analytics systems. The project involved handling multiple heterogeneous datasets collected from different cities and parking systems. Standardizing timestamp formats, occupancy structures, and spatial coordinates required careful preprocessing and iterative testing before forecasting and dashboard integration could be completed successfully.

The project also strengthened my collaborative development and communication skills. Working closely with machine learning forecasting workflows, dashboard development processes, and reporting activities provided valuable multidisciplinary project experience. Frequent collaboration between forecasting implementation, GIS analysis, and dashboard development helped ensure successful integration between all analytical components.

One of the most valuable aspects of the project was understanding how smart city technologies can support transportation management and urban mobility decision-making. The final smart parking intelligence platform demonstrated how machine learning forecasting, GIS hotspot mapping, API integration, and operational dashboards can work

together within a unified intelligent transportation framework.

If I were to repeat the project again, I would spend more time planning spatial visualization workflows, hotspot categorization standards, and dashboard integration structures earlier in the project lifecycle. More structured planning during the initial implementation stages could reduce later refinement activities and improve operational consistency.

Overall, the project provided highly valuable practical experience in predictive analytics, GIS visualization, machine learning workflows, and transportation analytics systems. It strengthened my technical knowledge, collaborative communication skills, and understanding of how intelligent technologies can support smart city transportation management applications.

2. Peer Review

My primary role in the project involved GIS-based hotspot analysis, spatial visualization support, dashboard testing, documentation preparation, and operational reporting assistance. I mainly worked on spatial mapping activities, occupancy categorization review, testing workflows, and report coordination. I also contributed to hotspot visualization refinement, dashboard validation, and GIS-based occupancy analysis activities.

Throughout the project, I supported the development of spatial intelligence components within the smart parking forecasting framework. This included assisting with hotspot mapping integration, reviewing occupancy categorization consistency, validating GIS visualization outputs, and supporting operational dashboard testing activities.

S M Amit Hasan mainly contributed to machine learning implementation, forecasting pipeline development, feature engineering, and occupancy prediction workflows. Their contribution was critical for forecasting model development, prediction optimization, API integration, and analytical validation activities.

Shovan Barai mainly handled Power BI dashboard development, forecasting visualization, KPI analytics, and operational reporting dashboards. Their contribution significantly improved the visual presentation quality and operational usability of the smart parking intelligence platform.

The group maintained collaborative coordination and regular communication throughout implementation, testing, dashboard integration, and report preparation phases. Frequent discussions and collaborative testing sessions helped ensure that forecasting outputs, hotspot visualizations, occupancy categories, and dashboard analytics remained synchronized throughout the project lifecycle.

One of the strongest aspects of the teamwork process was the integration of different technical domains into a unified operational framework. Since the project combined machine learning forecasting, GIS mapping, dashboard analytics, API integration, and

transportation analytics, close coordination between team members became extremely important. Collaborative debugging, iterative refinement, and shared testing activities significantly improved the overall quality of the final implementation.

Overall, the teamwork structure was effective and collaborative. Each member contributed according to their specialization area while also supporting integration, testing, and refinement activities across multiple project stages. The collaborative workflow significantly contributed to the successful completion of the final smart parking intelligence platform.